

DEPARTMENT OF INFORMATION TECHNOLOGY

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MODERN NETWORKING

For

Semester II

Submitted By:

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Msc.IT (Sem II)

Principal/Stamp

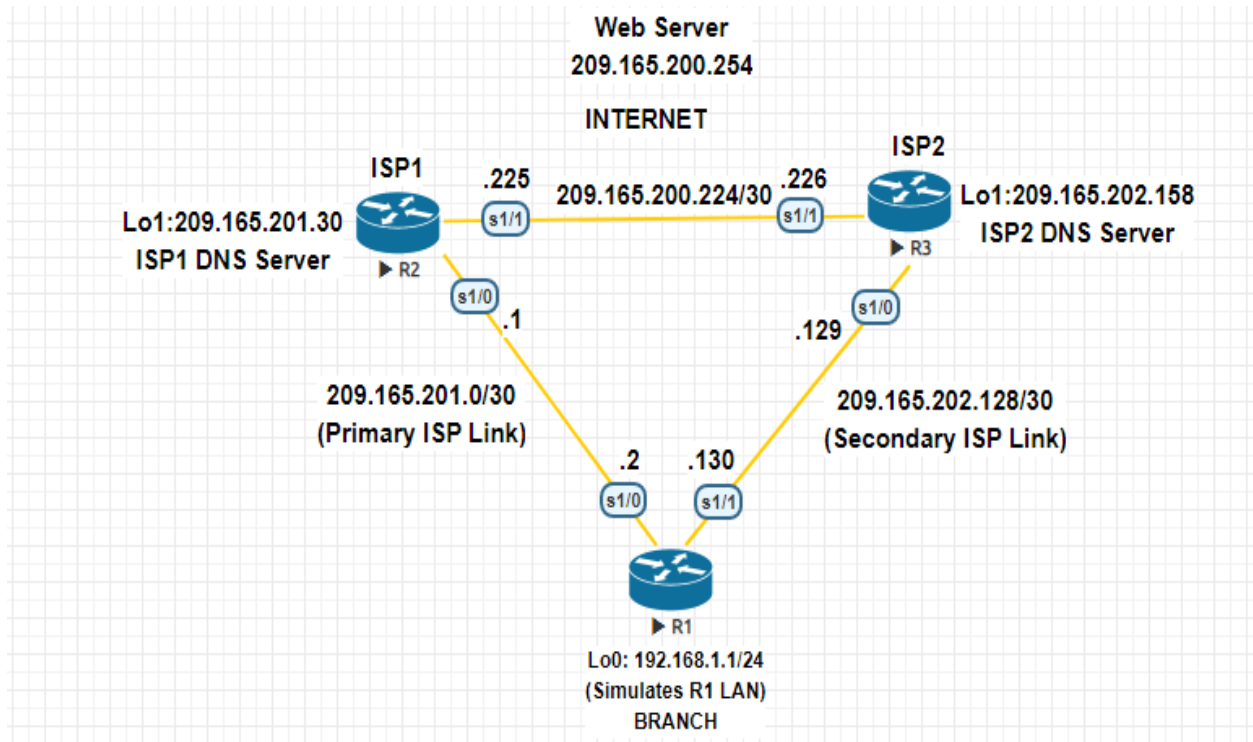
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Practical - 1

Configure IP SLA Tracking and Path Control Topology

NETWORK TOPOLOGY



TASKS

- Configure and verify the IP SLA feature.
- Test the IP SLA tracking feature.
- Verify the configuration and operation using show and debug commands

R1

Router>enable

Router#conf t

Router(config)#hostname R1

R1(config)#interface Loopback 0

R1(config-if)#ip address 192.168.1.1 255.255.255.0

R1(config-if)#exit

R1(config)#interface s1/0

R1(config-if)#ip address 209.165.201.2 255.255.255.252

R1(config-if)#no shutdown

R1(config-if)#exit

R1(config)#interface s1/1

R1(config-if)#ip address 209.165.202.130 255.255.255.252

R1(config-if)#no shutdown

R1(config-if)#exit

R1(config)#ip route 0.0.0.0 0.0.0.0 209.165.201.1

R1(config)#ip sla 12

R1(config-ip-sla)#icmp-echo 209.165.201.30

R1(config-ip-sla-echo)#frequency 11

R1(config-ip-sla-echo)#exit

R1(config)#ip sla schedule 12 life forever start-time now

R1#sh ip sla configuration 12

IP SLAs Infrastructure Engine-III

Entry number: 12

Owner:

Tag:

Operation timeout (milliseconds): 5000

Type of operation to perform: icmp-echo

Target address/Source address: 209.165.201.30/0.0.0.0

Type Of Service parameter: 0x0

Request size (ARR data portion): 28

Verify data: No

Vrf Name:

Schedule:

Operation frequency (seconds): 11 (not considered if randomly scheduled)

Next Scheduled Start Time: Start Time already passed

Group Scheduled : FALSE

Randomly Scheduled : FALSE

Life (seconds): Forever

Entry Ageout (seconds): never

Recurring (Starting Everyday): FALSE

Status of entry (SNMP RowStatus): Active

Threshold (milliseconds): 5000

Distribution Statistics:

Number of statistic hours kept: 2

Number of statistic distribution buckets kept: 1

Statistic distribution interval (milliseconds): 20

Enhanced History:

History Statistics:

Number of history Lives kept: 0

Number of history Buckets kept: 15

History Filter Type: None

R1#sh ip sla statistics

IPSLAs Latest Operation Statistics

IPSLA operation id: 12

Latest RTT: 11 milliseconds

Latest operation start time: 18:21:25 EET Thu Apr 9 2020

Latest operation return code: OK

Number of successes: 22

Number of failures: 0

Operation time to live: Forever

R1(config)#ip sla 24

R1(config-ip-sla)#icmp-echo 209.165.202.158

R1(config-ip-sla-echo)#frequency 10

R1(config-ip-sla-echo)#exit

R1(config)#ip sla schedule 24 life forever start-time now

R1#sh ip sla configuration 24

IP SLAs Infrastructure Engine-III

Entry number: 24

Owner:

Tag:

Operation timeout (milliseconds): 5000

Type of operation to perform: icmp-echo

Target address/Source address: 209.165.202.158/0.0.0.0

Type Of Service parameter: 0x0

Request size (ARR data portion): 28

Verify data: No

Vrf Name:

Schedule:

Operation frequency (seconds): 10 (not considered if randomly scheduled)

Next Scheduled Start Time: Start Time already passed

Group Scheduled : FALSE

Randomly Scheduled : FALSE

Life (seconds): Forever

Entry Ageout (seconds): never

Recurring (Starting Everyday): FALSE

Status of entry (SNMP RowStatus): Active

Threshold (milliseconds): 5000

Distribution Statistics:

Number of statistic hours kept: 2

Number of statistic distribution buckets kept: 1

Statistic distribution interval (milliseconds): 20

Enhanced History:

History Statistics:

Number of history Lives kept: 0

Number of history Buckets kept: 15

History Filter Type: None

R1#sh ip sla statistics 24

IPSLAs Latest Operation Statistics

IPSLA operation id: 24

Latest RTT: 20 milliseconds

Latest operation start time: 18:33:25 EET Thu Apr 9 2020

Latest operation return code: OK

Number of successes: 16

Number of failures: 0

Operation time to live: Forever

R1(config)#no ip route 0.0.0.0 0.0.0.0 209.165.201.1

R1(config)#ip route 0.0.0.0 0.0.0.0 209.165.201.1 5

R1#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP

a - application route

+ - replicated route, % - next hop override

Gateway of last resort is 209.165.201.1 to network 0.0.0.0

S* 0.0.0.0/0 [5/0] via 209.165.201.1

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.1.0/24 is directly connected, Loopback0

L 192.168.1.1/32 is directly connected, Loopback0

209.165.201.0/24 is variably subnetted, 2 subnets, 2 masks

C 209.165.201.0/30 is directly connected, Serial1/0

L 209.165.201.2/32 is directly connected, Serial1/0

209.165.202.0/24 is variably subnetted, 2 subnets, 2 masks

C 209.165.202.128/30 is directly connected, Serial1/1

L 209.165.202.130/32 is directly connected, Serial1/1

R1(config)#track 1 ip sla 12 reachability

R1(config-track)#delay down 10 up 1

R1(config-track)#exit

R1(config)#ip route 0.0.0.0 0.0.0.0 209.165.201.1 2 track 1

R1(config)#track 2 ip sla 12 reachability

R1(config-track)#delay down 10 up 1

R1(config-track)#exit

R1(config)#ip route 0.0.0.0 0.0.0.0 209.165.201.1 3 track 2

R1#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP

a - application route

+ - replicated route, % - next hop override

Gateway of last resort is 209.165.201.1 to network 0.0.0.0

S* 0.0.0.0/0 [3/0] via 209.165.201.1

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.1.0/24 is directly connected, Loopback0

L 192.168.1.1/32 is directly connected, Loopback0

209.165.201.0/24 is variably subnetted, 2 subnets, 2 masks

C 209.165.201.0/30 is directly connected, Serial1/0

L 209.165.201.2/32 is directly connected, Serial1/0

209.165.202.0/24 is variably subnetted, 2 subnets, 2 masks

C 209.165.202.128/30 is directly connected, Serial1/1

L 209.165.202.130/32 is directly connected, Serial1/1

R1#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP

a - application route

+ - replicated route, % - next hop override

Gateway of last resort is 209.165.201.1 to network 0.0.0.0

S* 0.0.0.0/0 [5/0] via 209.165.201.1

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.1.0/24 is directly connected, Loopback0

L 192.168.1.1/32 is directly connected, Loopback0

209.165.201.0/24 is variably subnetted, 2 subnets, 2 masks

C 209.165.201.0/30 is directly connected, Serial1/0

L 209.165.201.2/32 is directly connected, Serial1/0

209.165.202.0/24 is variably subnetted, 2 subnets, 2 masks

C 209.165.202.128/30 is directly connected, Serial1/1

L 209.165.202.130/32 is directly connected, Serial1/1

R1#sh ip sla statistics

IPSLAs Latest Operation Statistics

IPSLA operation id: 12

Latest RTT: NoConnection/Busy/Timeout

Latest operation start time: 19:02:29 EET Thu Apr 9 2020

Latest operation return code: Timeout

Number of successes: 227

Number of failures: 19

Operation time to live: Forever

IPSLA operation id: 24

Latest RTT: 20 milliseconds

Latest operation start time: 19:02:35 EET Thu Apr 9 2020

Latest operation return code: OK

Number of successes: 190

Number of failures: 1

Operation time to live: Forever

R1#trace 209.165.200.254 source 192.168.1.1

Type escape sequence to abort.

Tracing the route to 209.165.200.254

VRF info: (vrf in name/id, vrf out name/id)

1 209.165.201.1 10 msec 14 msec *

R1#sh ip sla statistics

IPSLAs Latest Operation Statistics

IPSLA operation id: 12

Latest RTT: 10 milliseconds

Latest operation start time: 19:07:04 EET Thu Apr 9 2020

Latest operation return code: OK

Number of successes: 236

Number of failures: 35

Operation time to live: Forever

IPSLA operation id: 24

Latest RTT: 21 milliseconds

Latest operation start time: 19:07:05 EET Thu Apr 9 2020

Latest operation return code: OK

Number of successes: 217

Number of failures: 1

Operation time to live: Forever

R1#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP

a - application route

+ - replicated route, % - next hop override

Gateway of last resort is 209.165.201.1 to network 0.0.0.0

S* 0.0.0.0/0 [3/0] via 209.165.201.1

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.1.0/24 is directly connected, Loopback0

L 192.168.1.1/32 is directly connected, Loopback0

209.165.201.0/24 is variably subnetted, 2 subnets, 2 masks

C 209.165.201.0/30 is directly connected, Serial1/0

L 209.165.201.2/32 is directly connected, Serial1/0

209.165.202.0/24 is variably subnetted, 2 subnets, 2 masks

C 209.165.202.128/30 is directly connected, Serial1/1

L 209.165.202.130/32 is directly connected, Serial1/1

ISP1 (R2)

Router>enable

Router#conf t

Router(config)#hostname ISP1

ISP1(config)#interface Loopback0

ISP1(config-if)#description Simulated Internet Web Server

ISP1(config-if)#ip address 209.165.200.254 255.255.255.255

ISP1(config-if)#exit

ISP1(config)#interface Loopback1

ISP1(config-if)#ip address 209.165.201.30 255.255.255.255

ISP1(config-if)#exit

ISP1(config)#interface s1/0

ISP1(config-if)#ip address 209.165.201.1 255.255.255.252

ISP1(config-if)#no shutdown

ISP1(config-if)#exit

ISP1(config)#interface s1/1

ISP1(config-if)#ip address 209.165.200.225 255.255.255.252

ISP1(config-if)#no shutdown

ISP1(config-if)#exit

```
ISP1(config)#router eigrp 200
ISP1(config-router)#network 209.165.200.224
ISP1(config-router)#network 209.165.201.0
ISP1(config-router)#no auto-summary
ISP1(config-router)#exit
ISP1(config)#ip route 192.168.1.0 255.255.255.0 209.165.201.2
ISP1(config)#interface loopback 1
ISP1(config-if)#shut
ISP1(config)#interface loopback 1
ISP1(config-if)#no shutdown
```

ISP2 (R3)

```
Router>enable
Router#conf t
Router(config)#hostname ISP2
ISP2(config)#interface Loopback0
ISP2(config-if)#description Simulated Internet Web Server
ISP2(config-if)#ip address 209.165.200.254 255.255.255.255
ISP2(config-if)#exit
ISP2(config)#interface Loopback1
ISP2(config-if)#ip address 209.165.202.158 255.255.255.255
ISP2(config-if)#exit
ISP2(config)#interface s1/1
ISP2(config-if)#ip address 209.165.200.226 255.255.255.252
```

ISP2(config-if)#no shutdown

ISP2(config-if)#exit

ISP2(config)#interface s1/0

ISP2(config-if)#ip address 209.165.202.129 255.255.255.252

ISP2(config-if)#no shutdown

ISP2(config-if)#exit

ISP2(config)#router eigrp 200

ISP2(config-router)#network 209.165.200.224

ISP2(config-router)#network 209.165.202.128

ISP2(config-router)#no auto-summary

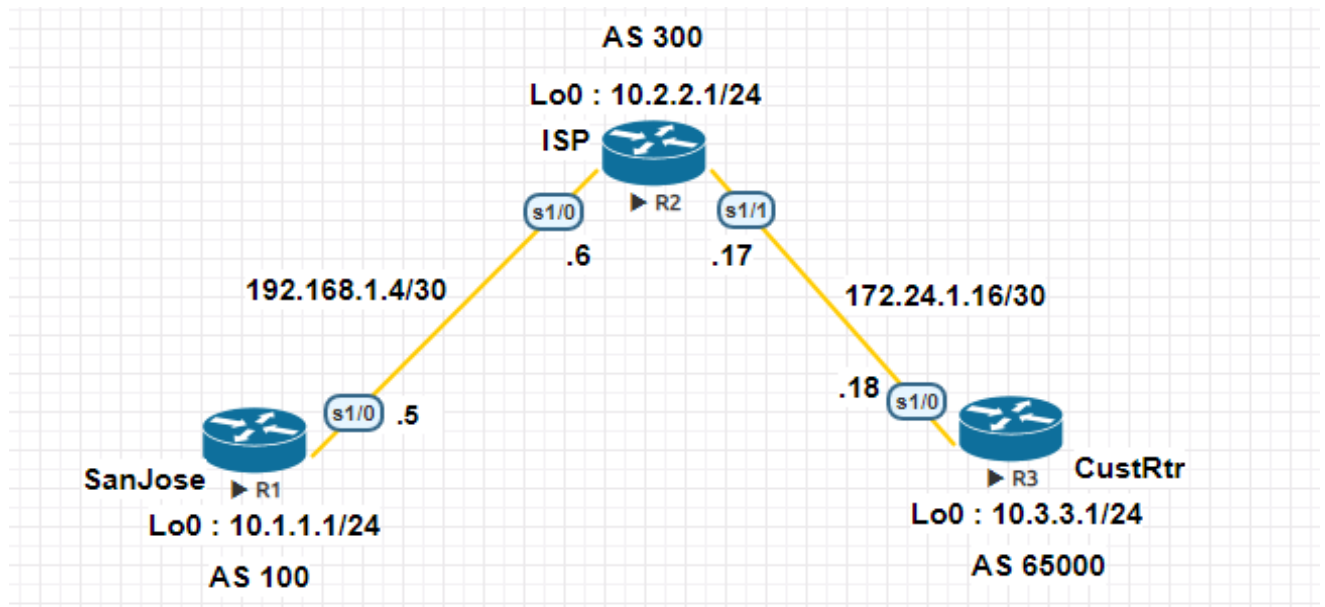
ISP2(config-router)#exit

ISP2(config)#ip route 192.168.1.0 255.255.255.0 209.165.202.130

Practical - 2

Using the AS_PATH Attribute

NETWORK TOPOLOGY



TASKS

- Use BGP commands to prevent private AS numbers from being advertised to the outside world.
- Use the AS_PATH attribute to filter BGP routes based on their source AS numbers.

SanJose

Router>enable

Router#conf t

Router(config)#hostname SanJose

SanJose(config)#interface Loopback0

SanJose(config-if)#ip address 10.1.1.1 255.255.255.0

SanJose(config-if)#exit

SanJose(config)#interface Serial1/0

SanJose(config-if)#ip address 192.168.1.5 255.255.255.252

SanJose(config-if)#no shutdown

SanJose(config-if)#end

SanJose(config)#router bgp 100

SanJose(config-router)#network 10.1.1.0 mask 255.255.255.0

SanJose(config-router)#neighbor 192.168.1.6 remote-as 300

SanJose(config-router)#exit

SanJose#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP

a - application route

+ - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks

C 10.1.1.0/24 is directly connected, Loopback0

L 10.1.1.1/32 is directly connected, Loopback0

B 10.2.2.0/24 [20/0] via 192.168.1.6, 00:05:47

B 10.3.3.0/24 [20/0] via 192.168.1.6, 00:02:13

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.1.4/30 is directly connected, Serial1/0

L 192.168.1.5/32 is directly connected, Serial1/0

SanJose#sh ip bgp

BGP table version is 4, local router ID is 10.1.1.1

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,

r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,

x best-external, a additional-path, c RIB-compressed,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.1.1.0/24	0.0.0.0	0	32768	i	

```
*> 10.2.2.0/24    192.168.1.6      0      0 300 i
*> 10.3.3.0/24    192.168.1.6      0 300 65000 i
```

SanJose#sh ip bgp

BGP table version is 5, local router ID is 10.1.1.1

**Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,**

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.1.1.0/24	0.0.0.0	0	32768	i	
*> 10.2.2.0/24	192.168.1.6	0	0	300	i
*> 10.3.3.0/24	192.168.1.6		0	300	i

ISP

Router>enable

Router#conf t

Router(config)#hostname ISP

ISP(config)#interface Loopback0

ISP(config-if)#ip address 10.2.2.1 255.255.255.0

ISP(config-if)#exit

ISP(config)#interface Serial1/0

ISP(config-if)#ip address 192.168.1.6 255.255.255.252

ISP(config-if)#no shutdown

ISP(config-if)#exit

ISP(config)#interface Serial1/1

ISP(config-if)#ip address 172.24.1.17 255.255.255.252

ISP(config-if)#no shutdown

ISP(config-if)#end

ISP(config)#router bgp 300

ISP(config-router)#network 10.2.2.0 mask 255.255.255.0

ISP(config-router)#neighbor 192.168.1.5 remote-as 100

ISP(config-router)#neighbor 172.24.1.18 remote-as 65000

ISP(config)#router bgp 300

ISP(config-router)#neighbor 192.168.1.5 remove-private-as

ISP(config-router)#end

ISP#clear ip bgp * soft

ISP(config)#ip as-path access-list 1 deny ^100\$

ISP(config)#ip as-path access-list 1 permit .*

ISP(config)#router bgp 300

ISP(config-router)#neighbor 172.24.1.18 filter-list 1 out

ISP(config-router)#end

ISP#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP

a - application route

+ - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks

B 10.1.1.0/24 [20/0] via 192.168.1.5, 00:46:41

C 10.2.2.0/24 is directly connected, Loopback0

L 10.2.2.1/32 is directly connected, Loopback0

B 10.3.3.0/24 [20/0] via 172.24.1.18, 00:43:07

172.24.0.0/16 is variably subnetted, 2 subnets, 2 masks

C 172.24.1.16/30 is directly connected, Serial1/1

L 172.24.1.17/32 is directly connected, Serial1/1

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.1.4/30 is directly connected, Serial1/0

L 192.168.1.6/32 is directly connected, Serial1/0

ISP#show ip bgp regexp ^100\$

BGP table version is 4, local router ID is 10.2.2.1

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,

r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,

x best-external, a additional-path, c RIB-compressed,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.1.1.0/24	192.168.1.5	0	0	100	i

CustRtr

Router>enable

Router#conf t

Router(config)#hostname CustRtr

CustRtr(config)#interface Loopback0

CustRtr(config-if)#ip address 10.3.3.1 255.255.255.0

CustRtr(config-if)#exit

CustRtr(config)#interface Serial1/0

CustRtr(config-if)#ip address 172.24.1.18 255.255.255.252

CustRtr(config-if)#no shutdown

CustRtr(config-if)#end

CustRtr(config)#router bgp 65000

CustRtr(config-router)#network 10.3.3.0 mask 255.255.255.0

CustRtr(config-router)#neighbor 172.24.1.17 remote-as 300

CustRtr(config-router)#end

CustRtr#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP

a - application route

+ - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks

B 10.2.2.0/24 [20/0] via 172.24.1.17, 00:45:59

C 10.3.3.0/24 is directly connected, Loopback0

L 10.3.3.1/32 is directly connected, Loopback0

172.24.0.0/16 is variably subnetted, 2 subnets, 2 masks

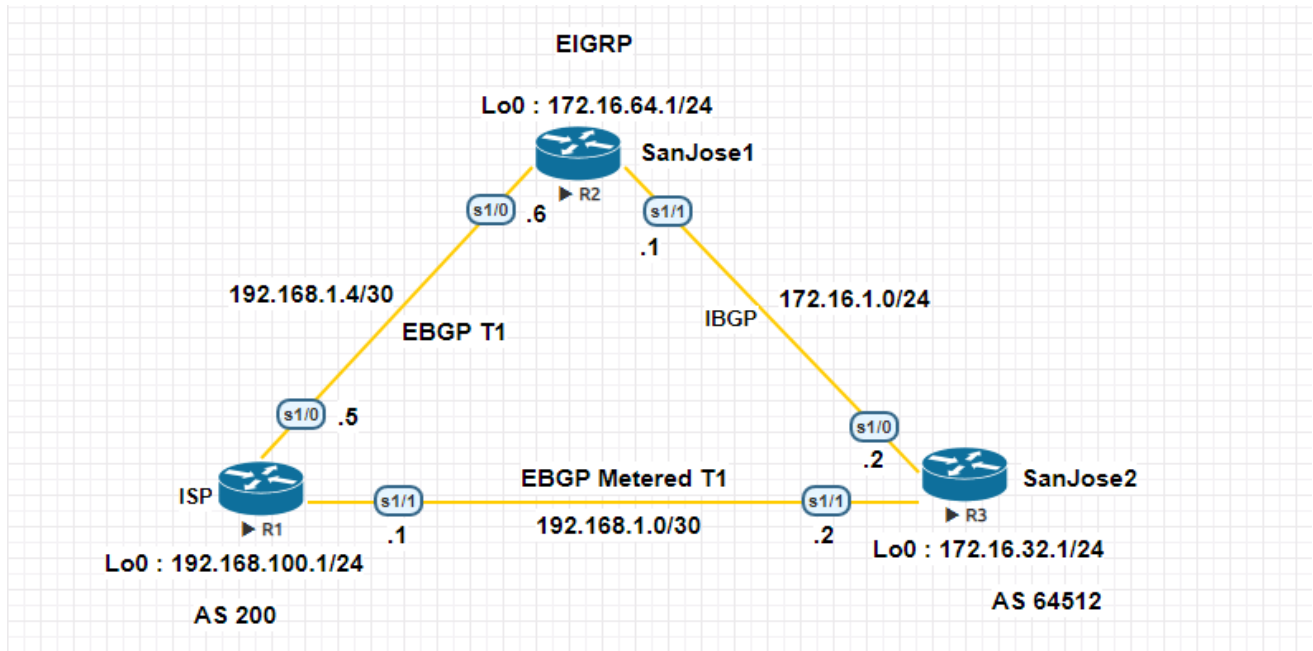
C 172.24.1.16/30 is directly connected, Serial1/0

L 172.24.1.18/32 is directly connected, Serial1/0

Practical - 3

Configuring IBGP and EBGP Sessions, Local Preference, and MED

NETWORK TOPOLOGY



TASKS

- For IBGP peers to correctly exchange routing information, use the next-hop-self command with the Local-Preference and MED attributes.
- Ensure that the flat-rate, unlimited-use T1 link is used for sending and receiving data to and from the AS 200 on ISP and that the metered T1 only be used in the event that the primary T1 link has failed

R1(ISP)

Router>enable

Router#conf t

Router(config)#hostname ISP

ISP(config)#interface Loopback0

ISP(config-if)#ip address 192.168.100.1 255.255.255.0

ISP(config-if)#exit

ISP(config)#interface Serial1/0

ISP(config-if)#ip address 192.168.1.5 255.255.255.252

ISP(config-if)#no shutdown

ISP(config-if)#exit

ISP(config)#interface Serial1/1

ISP(config-if)#ip address 192.168.1.1 255.255.255.252

ISP(config-if)#no shutdown

ISP(config-if)#exit

ISP(config)#router bgp 200

ISP(config-router)#network 192.168.100.0

ISP(config-router)#neighbor 192.168.1.6 remote-as 64512

ISP(config-router)#neighbor 192.168.1.2 remote-as 64512

ISP(config-router)#exit

ISP#sh ip bgp

BGP table version is 3, local router ID is 192.168.100.1

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,

r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,

x best-external, a additional-path, c RIB-compressed,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*	172.16.0.0	192.168.1.2	0	0	64512	i
*>		192.168.1.6	0	0	64512	i
*>	192.168.100.0	0.0.0.0	0		32768	i

ISP#ping 172.16.1.1 source 192.168.100.1

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 172.16.1.1, timeout is 2 seconds:

Packet sent with a source address of 192.168.100.1

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 10/10/11 ms

ISP#ping 172.16.32.1 source 192.168.100.1

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 172.16.32.1, timeout is 2 seconds:

Packet sent with a source address of 192.168.100.1

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 15/15/16 ms

ISP#ping 172.16.1.2 source 192.168.100.1

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 172.16.1.2, timeout is 2 seconds:

Packet sent with a source address of 192.168.100.1

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 15/17/25 ms

ISP(config)#router bgp 200

ISP(config-router)#network 192.168.1.0 mask 255.255.255.252

ISP(config-router)#network 192.168.1.4 mask 255.255.255.252

ISP(config-router)#exit

ISP#sh ip bgp

BGP table version is 5, local router ID is 192.168.100.1

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,

r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,

x best-external, a additional-path, c RIB-compressed,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*	172.16.0.0	192.168.1.6	0	0	64512	i
*>		192.168.1.2	0	0	64512	i
*>	192.168.1.0/30	0.0.0.0	0		32768	i

```
*> 192.168.1.4/30 0.0.0.0      0    32768 i
*> 192.168.100.0 0.0.0.0      0    32768 i
```

ISP#sh ip bgp

BGP table version is 6, local router ID is 192.168.100.1

**Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,**

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 172.16.0.0	192.168.1.6	50		0	64512 i
*	192.168.1.2	75		0	64512 i
*> 192.168.1.0/30	0.0.0.0	0		32768	i
*> 192.168.1.4/30	0.0.0.0	0		32768	i
*> 192.168.100.0	0.0.0.0	0		32768	i

ISP#ping 172.16.1.1

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 172.16.1.1, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 9/10/11 ms

ISP#ping 172.16.1.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 172.16.1.2, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 20/21/25 ms

ISP#traceroute 172.16.1.1

Type escape sequence to abort.

Tracing the route to 172.16.1.1

VRF info: (vrf in name/id, vrf out name/id)

1 192.168.1.6 10 msec 10 msec *

ISP#traceroute 172.16.1.2

Type escape sequence to abort.

Tracing the route to 172.16.1.2

VRF info: (vrf in name/id, vrf out name/id)

1 192.168.1.6 10 msec 10 msec 13 msec

2 172.16.1.2 [AS 64512] 20 msec 19 msec *

R2 (SanJose1)

Router>enable

Router#conf t

Router(config)#hostname SanJose1

SanJose1(config)#interface Loopback0

SanJose1(config-if)#ip address 172.16.64.1 255.255.255.0

SanJose1(config-if)#ip address 172.16.64.1 255.255.255.0

SanJose1(config-if)#exit

SanJose1(config)#interface Serial1/0

SanJose1(config-if)#ip address 192.168.1.6 255.255.255.252

SanJose1(config-if)#no shutdown

SanJose1(config-if)#exit

SanJose1(config)#interface Serial1/1

SanJose1(config-if)#ip address 172.16.1.1 255.255.255.0

SanJose1(config-if)#no shutdown

SanJose1(config-if)#exit

SanJose1(config)#router eigrp 64512

SanJose1(config-router)#network 172.16.0.0

SanJose1(config-router)#no auto-summary

SanJose1(config-router)#exit

SanJose1(config)#router bgp 64512

SanJose1(config-router)#neighbor 172.16.32.1 remote-as 64512

SanJose1(config-router)#neighbor 172.16.32.1 update-source loopback0

SanJose1(config-router)#exit

```

SanJose1(config)#ip route 172.16.0.0 255.255.0.0 null 0
SanJose1(config)#router bgp 64512
SanJose1(config-router)#network 172.16.0.0
SanJose1(config-router)#neighbor 192.168.1.5 remote-as 200
SanJose1(config-router)#exit
SanJose1(config)#router bgp 64512
SanJose1(config-router)#neighbor 172.16.32.1 next-hop-self
SanJose1(config-router)#exit

```

```
SanJose1#sh ip bgp
```

BGP table version is 5, local router ID is 172.16.64.1

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

Network	Next Hop	Metric	LocPrf	Weight	Path
* i 172.16.0.0	172.16.32.1	0	100	0	i
*>	0.0.0.0	0	32768		i
* i 192.168.1.0/30	172.16.32.1	0	100	0	200 i
*>	192.168.1.5	0	0	200	i
r i 192.168.1.4/30	172.16.32.1	0	100	0	200 i
r>	192.168.1.5	0	0	200	i

```
* i 192.168.100.0 172.16.32.1 0 100 0 200 i
*> 192.168.1.5 0 0 200 i
```

```
SanJose1(config)#route-map PRIMARY_T1_IN permit 10
SanJose1(config-route-map)#set local-preference 160
SanJose1(config-route-map)#exit
SanJose1(config)#router bgp 64512
SanJose1(config-router)#neighbor 192.168.1.5 route-map PRIMARY_T1_IN in
SanJose1(config-router)#exit
SanJose1#clear ip bgp * soft
```

```
SanJose1#sh ip bgp
BGP table version is 8, local router ID is 172.16.64.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found
```

Network	Next Hop	Metric	LocPrf	Weight	Path
* i 172.16.0.0	172.16.32.1	0	100	0	i
*> 0.0.0.0	0	32768			i
*> 192.168.1.0/30	192.168.1.5	0	160	0	200 i
r> 192.168.1.4/30	192.168.1.5	0	160	0	200 i

```
*> 192.168.100.0 192.168.1.5 0 160 0 200 i
```

```
SanJose1(config)#route-map PRIMARY_T1_MED_OUT permit 10
```

```
SanJose1(config-route-map)#set Metric 50
```

```
SanJose1(config-route-map)#exit
```

```
SanJose1(config)#router bgp 64512
```

```
SanJose1(config-router)#neighbor 192.168.1.5 route-map  
PRIMARY_T1_MED_OUT out
```

```
SanJose1(config-router)#exit
```

```
SanJose1(config)#exit
```

```
SanJose1#clear ip bgp * soft
```

```
SanJose1#sh ip bgp
```

```
BGP table version is 8, local router ID is 172.16.64.1
```

```
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
```

```
          r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
```

```
          x best-external, a additional-path, c RIB-compressed,
```

```
Origin codes: i - IGP, e - EGP, ? - incomplete
```

```
RPKI validation codes: V valid, I invalid, N Not found
```

Network	Next Hop	Metric	LocPrf	Weight	Path
* i 172.16.0.0	172.16.32.1	0	100	0	i
*>	0.0.0.0	0	32768		i
*> 192.168.1.0/30	192.168.1.5	0	160	0	200 i
r> 192.168.1.4/30	192.168.1.5	0	160	0	200 i

```
*> 192.168.100.0 192.168.1.5 0 160 0 200 i
```

SanJose1#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP

a - application route

+ - replicated route, % - next hop override

Gateway of last resort is not set

172.16.0.0/16 is variably subnetted, 6 subnets, 3 masks

S 172.16.0.0/16 is directly connected, Null0

C 172.16.1.0/24 is directly connected, Serial1/1

L 172.16.1.1/32 is directly connected, Serial1/1

D 172.16.32.0/24 [90/2297856] via 172.16.1.2, 01:28:25, Serial1/1

C 172.16.64.0/24 is directly connected, Loopback0

L 172.16.64.1/32 is directly connected, Loopback0

192.168.1.0/24 is variably subnetted, 3 subnets, 2 masks

B 192.168.1.0/30 [20/0] via 192.168.1.5, 00:45:28

C 192.168.1.4/30 is directly connected, Serial1/0
L 192.168.1.6/32 is directly connected, Serial1/0
B 192.168.100.0/24 [20/0] via 192.168.1.5, 00:45:28

After issuing ip default-network

SanJose1(config)#ip default-network 192.168.100.0

SanJose1(config)#end

SanJose1#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP

a - application route

+ - replicated route, % - next hop override

Gateway of last resort is 192.168.1.5 to network 192.168.100.0

S* 0.0.0.0/0 [20/0] via 192.168.1.5

172.16.0.0/16 is variably subnetted, 6 subnets, 3 masks

S 172.16.0.0/16 is directly connected, Null0

C 172.16.1.0/24 is directly connected, Serial1/1
L 172.16.1.1/32 is directly connected, Serial1/1
D 172.16.32.0/24 [90/2297856] via 172.16.1.2, 01:33:38, Serial1/1
C 172.16.64.0/24 is directly connected, Loopback0
L 172.16.64.1/32 is directly connected, Loopback0
192.168.1.0/24 is variably subnetted, 3 subnets, 2 masks
B 192.168.1.0/30 [20/0] via 192.168.1.5, 00:50:41
C 192.168.1.4/30 is directly connected, Serial1/0
L 192.168.1.6/32 is directly connected, Serial1/0
B* 192.168.100.0/24 [20/0] via 192.168.1.5, 00:50:41

SanJose1#ping 192.168.1.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.1.2, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 14/15/16 ms

SanJose1#traceroute 192.168.1.2

Type escape sequence to abort.

Tracing the route to 192.168.1.2

VRF info: (vrf in name/id, vrf out name/id)

1 192.168.1.5 [AS 200] 10 msec 10 msec 10 msec

2 192.168.1.2 [AS 200] 15 msec 15 msec *

SanJose1#ping 192.168.1.1

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 9/9/11 ms

SanJose1#traceroute 192.168.1.1

Type escape sequence to abort.

Tracing the route to 192.168.1.1

VRF info: (vrf in name/id, vrf out name/id)

1 192.168.1.5 [AS 200] 10 msec 11 msec *

R3 (SanJose2)

Router>en

Router#conf t

Router(config)#hostname SanJose2

SanJose2(config)#interface Loopback0

SanJose2(config-if)#ip address 172.16.32.1 255.255.255.0

SanJose2(config-if)#exit

SanJose2(config)#interface Serial1/1

SanJose2(config-if)#ip address 192.168.1.2 255.255.255.252

SanJose2(config-if)#no shutdown

SanJose2(config-if)#exit

SanJose2(config)#interface Serial1/0

SanJose2(config-if)#ip address 172.16.1.2 255.255.255.0

```
SanJose2(config-if)#no shutdown
SanJose2(config-if)#exit
SanJose2(config)#router eigrp 64512
SanJose2(config-router)#network 172.16.0.0
SanJose2(config-router)#no auto-summary
SanJose2(config-router)#exit
SanJose2(config)#router bgp 64512
SanJose2(config-router)#neighbor 172.16.64.1 remote-as 64512
SanJose2(config-router)#neighbor 172.16.64.1 update-source loopback0
SanJose2(config-router)#exit
SanJose2(config)#ip route 172.16.0.0 255.255.0.0 null 0
SanJose2(config)#router bgp 64512
SanJose2(config-router)#network 172.16.0.0
SanJose2(config-router)#neighbor 192.168.1.1 remote-as 200
SanJose2(config-router)#exit

SanJose2#sh ip bgp summary
BGP router identifier 172.16.32.1, local AS number 64512
BGP table version is 4, main routing table version 4
2 network entries using 280 bytes of memory
4 path entries using 320 bytes of memory
4/2 BGP path/bestpath attribute entries using 576 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
```

0 BGP filter-list cache entries using 0 bytes of memory

BGP using 1200 total bytes of memory

BGP activity 2/0 prefixes, 4/0 paths, scan interval 60 secs

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down
----------	---	----	---------	---------	--------	-----	------	---------

State/PfxRcd

172.16.64.1	4	64512	31	32	4	0	0	00:24:41	2
-------------	---	-------	----	----	---	---	---	----------	---

192.168.1.1	4	200	8	6	4	0	0	00:01:22	1
-------------	---	-----	---	---	---	---	---	----------	---

SanJose2#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP

a - application route

+ - replicated route, % - next hop override

Gateway of last resort is not set

172.16.0.0/16 is variably subnetted, 6 subnets, 3 masks

S 172.16.0.0/16 is directly connected, Null0

C 172.16.1.0/24 is directly connected, Serial1/0

L 172.16.1.2/32 is directly connected, Serial1/0

- C 172.16.32.0/24 is directly connected, Loopback0**
- L 172.16.32.1/32 is directly connected, Loopback0**
- D 172.16.64.0/24 [90/2297856] via 172.16.1.1, 00:08:46, Serial1/0**
192.168.1.0/24 is variably subnetted, 3 subnets, 2 masks
- C 192.168.1.0/30 is directly connected, Serial1/1**
- L 192.168.1.2/32 is directly connected, Serial1/1**
- B 192.168.1.4/30 [20/0] via 192.168.1.1, 00:02:19**
- B 192.168.100.0/24 [20/0] via 192.168.1.1, 00:07:40**

SanJose2#sh ip bgp

BGP table version is 5, local router ID is 172.16.32.1

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

Network	Next Hop	Metric	LocPrf	Weight	Path
* i 172.16.0.0	172.16.64.1	0	100	0	i
*>	0.0.0.0	0	32768		i
r i 192.168.1.0/30	192.168.1.5	0	100	0	200 i
r>	192.168.1.1	0		0	200 i
* i 192.168.1.4/30	192.168.1.5	0	100	0	200 i
*>	192.168.1.1	0		0	200 i

```
* i 192.168.100.0 192.168.1.5 0 100 0 200 i
*> 192.168.1.1 0 0 200 i
```

SanJose2(config)#router bgp 64512

SanJose2(config-router)#neighbor 172.16.64.1 next-hop-self

SanJose2(config-router)#exit

SanJose2#sh ip bgp

BGP table version is 5, local router ID is 172.16.32.1

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,

r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,

x best-external, a additional-path, c RIB-compressed,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

Network	Next Hop	Metric	LocPrf	Weight	Path
* i 172.16.0.0	172.16.64.1	0	100	0	i
*>	0.0.0.0	0	32768		i
r i 192.168.1.0/30	172.16.64.1	0	100	0	200 i
r>	192.168.1.1	0	0	200	i
* i 192.168.1.4/30	172.16.64.1	0	100	0	200 i
*>	192.168.1.1	0	0	200	i
* i 192.168.100.0	172.16.64.1	0	100	0	200 i
*>	192.168.1.1	0	0	200	i

```

SanJose2(config)#route-map SECONDARY_T1_IN permit 10
SanJose2(config-route-map)#set local-preference 125
SanJose2(config-route-map)#exit
SanJose2(config)#router bgp 64512
SanJose2(config-router)#neighbor 192.168.1.1 route-map SECONDARY_T1_IN in
SanJose2(config-router)#exit
SanJose2#clear ip bgp * soft
SanJose2#sh ip bgp

BGP table version is 8, local router ID is 172.16.32.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

```

Network	Next Hop	Metric	LocPrf	Weight	Path
* i 172.16.0.0	172.16.64.1	0	100	0	i
*>	0.0.0.0	0	32768		i
r>i 192.168.1.0/30	172.16.64.1	0	160	0	200 i
r	192.168.1.1	0	125	0	200 i
*>i 192.168.1.4/30	172.16.64.1	0	160	0	200 i
*	192.168.1.1	0	125	0	200 i
*>i 192.168.100.0	172.16.64.1	0	160	0	200 i
*	192.168.1.1	0	125	0	200 i

```

SanJose2(config)#route-map SECONDARY_T1_MED_OUT permit 10
SanJose2(config-route-map)#set Metric 75
SanJose2(config-route-map)#exit
SanJose2(config)#router bgp 64512
SanJose2(config-router)#2.168.1.1 route-map SECONDARY_T1_MED_OUT out
SanJose2(config-router)#end
SanJose2#clear ip bgp * soft
SanJose2#sh ip bgp

BGP table version is 8, local router ID is 172.16.32.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,

Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

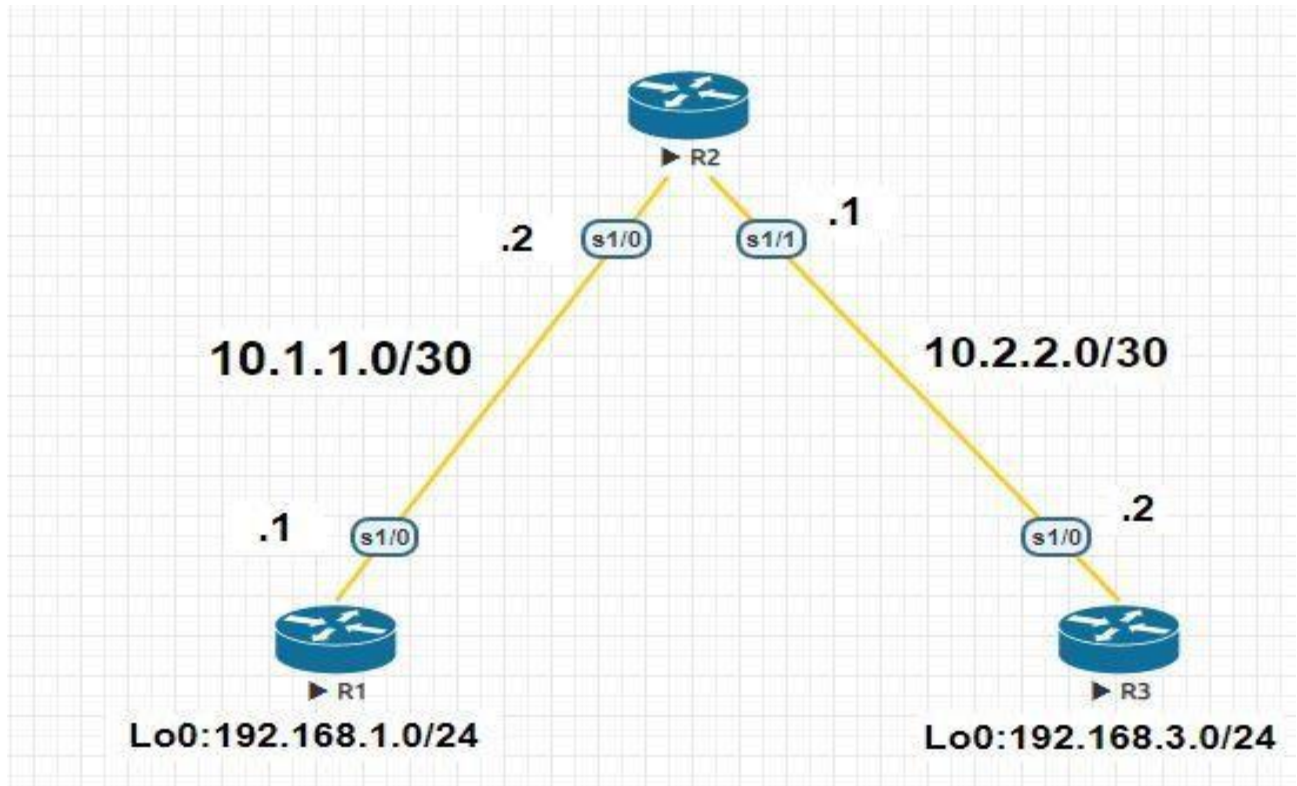
  Network      Next Hop      Metric LocPrf Weight Path
* i 172.16.0.0   172.16.64.1      0  100   0 i
*>           0.0.0.0      0  32768 i
r>i 192.168.1.0/30 172.16.64.1      0  160   0 200 i
r           192.168.1.1      0  125   0 200 i
*>i 192.168.1.4/30 172.16.64.1      0  160   0 200 i
*           192.168.1.1      0  125   0 200 i
*>i 192.168.100.0 172.16.64.1      0  160   0 200 i
*           192.168.1.1      0  125   0 200 i

```

Practical - 4

Secure the Management Plane

NETWORK TOPOLOGY



TASKS

- Secure Management Access
- Configure enhanced username password security
- Enable AAA RADIUS authentication
- Enable Secure Remote Management

R1

Router>en

Router#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#hostname R1

R1(config)#interface Loopback 0

*Dec 19 07:53:42.473: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

R1(config-if)#ip address 192.168.1.1 255.255.255.0

R1(config-if)#exit

R1(config)#interface s1/0

R1(config-if)#ip address 10.1.1.1 255.255.255.252

R1(config-if)#no shutdown

*Dec 19 07:57:21.998: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up

*Dec 19 07:57:22.999: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up

R1(config-if)#exit

R1(config)#exit

Configure static routes

- a. On R1, configure a default static route to ISP.

```
R1(config)# ip route 0.0.0.0 0.0.0.0 10.1.1.2
```

```
R1#show ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2 E1 - OSPF external type 1, E2 - OSPF external type 2 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP

a - application route

+ - replicated route, % - next hop override

Gateway of last resort is 10.1.1.2 to network 0.0.0.0

```
S* 0.0.0.0/0 [1/0] via 10.1.1.2
```

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

```
C 10.1.1.0/30 is directly connected, Serial1/0
```

```
L 10.1.1.1/32 is directly connected, Serial1/0
```

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks

```
C 192.168.1.0/24 is directly connected, Loopback0
```

```
L 192.168.1.1/32 is directly connected, Loopback0
```

Secure management access

R1(config)#security passwords min-length 10

R1(config)#enable secret class12345

R1(config)#line console 0

R1(config-line)#password ciscoconpass

R1(config-line)#exec-timeout 5 0

R1(config-line)#login

R1(config-line)#logging synchronous

R1(config-line)#exit

R1(config)#line vty 0 4

R1(config-line)#password ciscovtypass

R1(config-line)#exec-timeout 5 0

R1(config-line)#login

R1(config-line)#exit

R1(config)#line aux 0

R1(config-line)#no exec

R1(config-line)#end

R1(config)#service password-encryption

R1(config)#banner motd \$Unauthorized access strictly prohibited!\$

R1(config)#exit

Configure enhanced username password security

```
R1(config)#username JR-ADMIN secret class12345
```

```
R1(config)#username ADMIN secret class54321
```

```
R1(config)#line console 0
```

```
R1(config-line)#login local
```

```
R1(config-line)#end
```

```
R1(config)#line vty 0 4
```

```
R1(config-line)#login local
```

```
R1(config-line)#end
```

Enabling AAA RADIUS Authentication with Local User for Backup

```
R1(config)# aaa new-model
```

```
R1(config)# radius server RADIUS-1
```

```
R1(config-radius-server)# address ipv4 192.168.1.101
```

```
R1(config-radius-server)# key RADIUS-1-pa55w0rd
```

```
R1(config-radius-server)# exit
```

```
R1(config)# radius server RADIUS-2
```

```
R1(config-radius-server)# address ipv4 192.168.1.102
```

```
R1(config-radius-server)# key RADIUS-2-pa55w0rd
```

```
R1(config-radius-server)# exit
```

```
R1(config)# aaa group server radius RADIUS-GROUP
```

```
R1(config-sg-radius)# server name RADIUS-1
```

```
R1(config-sg-radius)# server name RADIUS-2
```

```
R1(config-sg-radius)# exit
```

```
R1(config)# aaa authentication login default group RADIUS-GROUP local
```

```
R1(config)# aaa authentication login TELNET-LOGIN group RADIUS-GROUP local-  
case
```

```
R1(config)# line vty 0 4
```

```
R1(config-line)# login authentication TELNET-LOGIN
```

R1(config-line)# exit

R2

Router>enable

Router#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#hostname R2

R2(config)#interface s1/0

R2(config-if)#ip address 10.1.1.2 255.255.255.252

R2(config-if)#no shutdown

***Dec 19 08:01:10.279: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up**

***Dec 19 08:01:11.279: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up**

R2(config-if)#exit

R2(config)#interface s1/1

R2(config-if)#ip address 10.2.2.1 255.255.255.252

R2(config-if)#no shutdown

***Dec 19 08:02:33.002: %LINK-3-UPDOWN: Interface Serial1/1, changed state to up**

***Dec 19 08:02:34.009: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/1, changed state to up**

R2(config-if)#exit

R2(config)#exit

Configure static routes

- a. On R2, configure two static routes.

```
R2(config)# ip route 192.168.1.0 255.255.255.0 10.1.1.1
```

```
R2(config)# ip route 192.168.3.0 255.255.255.0 10.2.2.2
```

```
R2#show ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B – BGP D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2 E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2 ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP

a - application route

+ - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks

C 10.1.1.0/30 is directly connected, Serial1/0

L 10.1.1.2/32 is directly connected, Serial1/0

C 10.2.2.0/30 is directly connected, Serial1/1

L 10.2.2.1/32 is directly connected, Serial1/1

S 192.168.1.0/24 [1/0] via 10.1.1.1

S 192.168.3.0/24 [1/0] via 10.2.2.2

Secure management access

R2(config)#security passwords min-length 10

R2(config)#enable secret class12345

R2(config)#line console 0

R2(config-line)#password ciscoconpass

R2(config-line)#exec-timeout 5 0

R2(config-line)#login

R2(config-line)#logging synchronous

R2(config-line)#exit

R2(config)#line vty 0 4

R2(config-line)#password ciscovtypass

R2(config-line)#exec-timeout 5 0

R2(config-line)#login

R2(config-line)#exit

R2(config)#line aux 0

R2(config-line)#no exec

R2(config-line)#end

R2(config)#service password-encryption

R2(config)#banner motd \$Unauthorized access strictly prohibited!\$

R2(config)#exit

Configure enhanced username password security

```
R2(config)#username JR-ADMIN secret class12345
```

```
R2(config)#username ADMIN secret class54321
```

```
R2(config)#line console 0
```

```
R2(config-line)#login local
```

```
R2(config-line)#end
```

```
R2(config)#line vty 0 4
```

```
R2(config-line)#login local
```

```
R2(config-line)#end
```

Enabling AAA RADIUS Authentication with Local User for Backup

```
R2(config)# aaa new-model
```

```
R2(config)# radius server RADIUS-1
```

```
R2(config-radius-server)# address ipv4 192.168.1.101
```

```
R2(config-radius-server)# key RADIUS-1-pa55w0rd
```

```
R2(config-radius-server)# exit
```

```
R2(config)# radius server RADIUS-2
```

```
R2(config-radius-server)# address ipv4 192.168.1.102
```

```
R2(config-radius-server)# key RADIUS-2-pa55w0rd
```

```
R2(config-radius-server)# exit
```

```
R2(config)# aaa group server radius RADIUS-GROUP
```

```
R2(config-sg-radius)# server name RADIUS-1
```

```
R2(config-sg-radius)# server name RADIUS-2
```

```
R2(config-sg-radius)# exit
```

```
R2(config)# aaa authentication login default group RADIUS-GROUP local
```

```
R2(config)# aaa authentication login TELNET-LOGIN group RADIUS-GROUP local-  
case
```

```
R2(config)# line vty 0 4
```

```
R2(config-line)# login authentication TELNET-LOGIN
```

R2(config-line)# exit

R3

Router>enable

Router#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#hostname R3

R3(config)#interface loopback 0

***Dec 19 08:07:50.079: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up**

R3(config-if)#ip address 192.168.3.1 255.255.255.0

R3(config-if)#exit

R3(config)#interface s1/0

R3(config-if)#ip address 10.2.2.2 255.255.255.252

R3(config-if)#no shutdown

R3(config-if)#exit

***Dec 19 08:09:26.986: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up**

***Dec 19 08:09:27.996: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up**

R3(config)#end

Configure static routes

- a. On R3, configure a default static route to ISP.**

R3(config)# ip route 0.0.0.0 0.0.0.0 10.2.2.1

R3#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B – BGP D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2 E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP

a - application route

+ - replicated route, % - next hop override

Gateway of last resort is 10.2.2.1 to network 0.0.0.0

S* 0.0.0.0/0 [1/0] via 10.2.2.1

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

C 10.2.2.0/30 is directly connected, Serial1/0

L 10.2.2.2/32 is directly connected, Serial1/0

192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.3.0/24 is directly connected, Loopback0

L 192.168.3.1/32 is directly connected, Loopback0

Secure management access

R3(config)#security passwords min-length 10

R3(config)#enable secret class12345

R3(config)#line console 0

R3(config-line)#password ciscoconpass

```
R3(config-line)#exec-timeout 5 0
R3(config-line)#login
R3(config-line)#logging synchronous
R3(config-line)#exit
R3(config)#line vty 0 4
R3(config-line)#password ciscovtypass
R3(config-line)#exec-timeout 5 0
R3(config-line)#login
R3(config-line)#exit
R3(config)#line aux 0
R3(config-line)#no exec
R3(config-line)#end
R3(config)#service password-encryption
R3(config)#banner motd $Unauthorized access strictly prohibited!$
```

Configure enhanced username password security

```
R3(config)#username JR-ADMIN secret class12345
R3(config)#username ADMIN secret class54321
R3(config)#line console 0
R3(config-line)#login local
R3(config-line)#exit
R3(config)#line vty 0 4
R3(config-line)#login local
R3(config-line)#exit
```

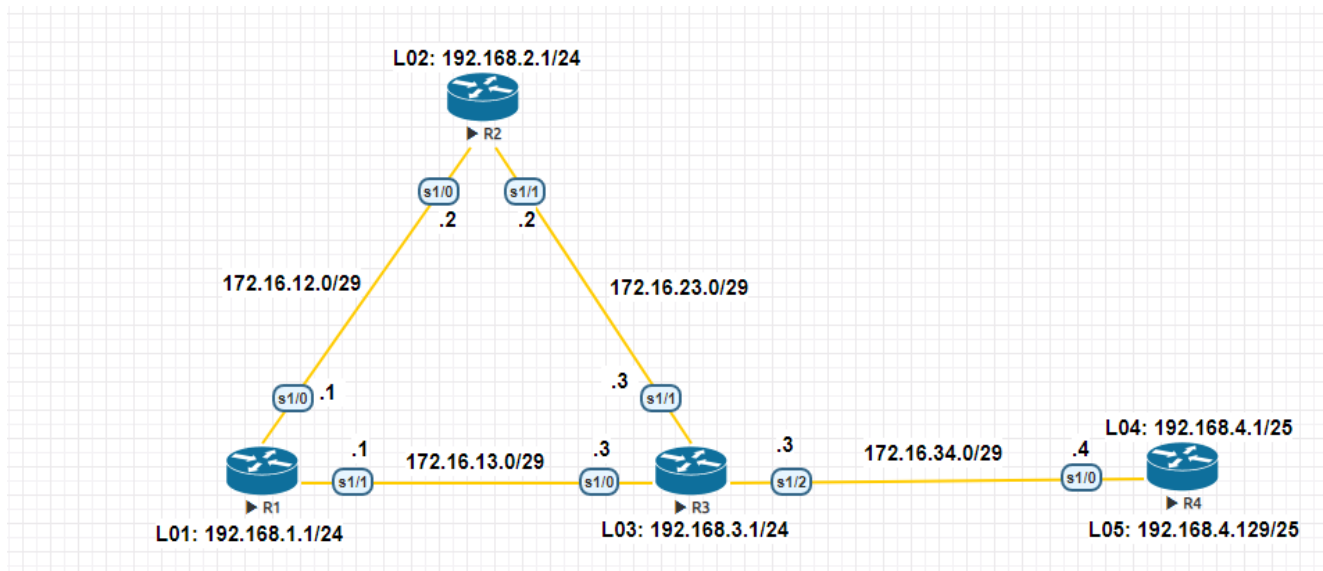
Enabling AAA RADIUS Authentication with Local User for Backup

```
R3(config)# aaa new-model
R3(config)# radius server RADIUS-1
R3(config-radius-server)# address ipv4 192.168.1.101
R3(config-radius-server)# key RADIUS-1-pa55w0rd
R3(config-radius-server)# exit
R3(config)# radius server RADIUS-2
R3(config-radius-server)# address ipv4 192.168.1.102
R3(config-radius-server)# key RADIUS-2-pa55w0rd
R3(config-radius-server)# exit
R3(config)# aaa group server radius RADIUS-GROUP
R3(config-sg-radius)# server name RADIUS-1
R3(config-sg-radius)# server name RADIUS-2
R3(config-sg-radius)# exit
R3(config)# aaa authentication login default group RADIUS-GROUP local
R3(config)# aaa authentication login TELNET-LOGIN group RADIUS-GROUP local-
case
R3(config)# line vty 0 4
R3(config-line)# login authentication TELNET-LOGIN
R3(config-line)# exit
```

Practical - 5

Configure and Verify Path Control Using PBR

NETWORK TOPOLOGY



TASKS

- Configure and verify policy-based routing.
- Select the required tools and commands to configure policy-based routing operations.
- Verify the configuration and operation by using the proper show and debug commands

R1

Router>enable

Router#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#hostname R1

R1(config)#interface Lo1

R1(config-if)#ip address 192.168.1.1 255.255.255.0

R1(config-if)#exit

R1(config)#interface s1/0

R1(config-if)#ip address 172.16.12.1 255.255.255.248

R1(config-if)#no shutdown

R1(config-if)#exit

R1(config)#interface s1/1

R1(config-if)#ip address 172.16.13.1 255.255.255.248

R1(config-if)#no shutdown

R1(config-if)#exit

R1(config)#router eigrp 100

R1(config-router)#network 192.168.1.0

R1(config-router)#network 172.16.12.0

R1(config-router)#network 172.16.13.0

R1(config-router)#no auto-summary

R1(config-router)#exit

R1#sh ip eigrp neighbors

EIGRP-IPv4 Neighbors for AS(100)

H	Address	Interface	Hold	Uptime	SRTT	RTO	Q	Seq
		(sec)	(ms)	Cnt	Num			
1	172.16.13.3	Se1/1	14	00:04:43	11	100	0	10
0	172.16.12.2	Se1/0	12	00:07:05	19	114	0	8

R1#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP

a - application route

+ - replicated route, % - next hop override

Gateway of last resort is not set

172.16.0.0/16 is variably subnetted, 6 subnets, 2 masks

C 172.16.12.0/29 is directly connected, Serial1/0

L 172.16.12.1/32 is directly connected, Serial1/0

C 172.16.13.0/29 is directly connected, Serial1/1

L 172.16.13.1/32 is directly connected, Serial1/1

- D 172.16.23.0/29 [90/2681856] via 172.16.13.3, 00:08:31, Serial1/1
[90/2681856] via 172.16.12.2, 00:08:31, Serial1/0
- D 172.16.34.0/29 [90/2681856] via 172.16.13.3, 00:08:31, Serial1/1
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
- C 192.168.1.0/24 is directly connected, Loopback1
- L 192.168.1.1/32 is directly connected, Loopback1
- D 192.168.2.0/24 [90/2297856] via 172.16.12.2, 00:08:31, Serial1/0
- D 192.168.3.0/24 [90/2297856] via 172.16.13.3, 00:08:31, Serial1/1
192.168.4.0/25 is subnetted, 2 subnets
- D 192.168.4.0 [90/2809856] via 172.16.13.3, 00:05:15, Serial1/1
- D 192.168.4.128 [90/2809856] via 172.16.13.3, 00:05:15, Serial1/1

R2

Router>enable

Router#conf t

Router(config)#hostname R2

R2(config)#interface Lo2

R2(config-if)#ip address 192.168.2.1 255.255.255.0

R2(config-if)#exit

R2(config)#interface s1/0

R2(config-if)#ip address 172.16.12.2 255.255.255.248

R2(config-if)#no shutdown

R2(config-if)#exit

R2(config)#interface s1/1

R2(config-if)#ip address 172.16.23.2 255.255.255.248

R2(config-if)#no shutdown

R2(config-if)#exit

R2(config)#router eigrp 100

R2(config-router)#network 192.168.2.0

R2(config-router)#network 172.16.12.0

R2(config-router)#network 172.16.23.0

R2(config-router)#no auto-summary

R2#sh ip eigrp neighbors

EIGRP-IPv4 Neighbors for AS(100)

H	Address	Interface	Hold Uptime	SRTT	RTO	Q	Seq
		(sec)	(ms) Cnt Num				
1	172.16.23.3	Se1/1	12 00:05:23	12	100	0	11
0	172.16.12.1	Se1/0	12 00:07:45	22	132	0	8

R3

Router>enable

Router#conf t

Router(config)#hostname R3

R3(config)#interface Lo3

R3(config-if)#ip address 192.168.3.1 255.255.255.0

R3(config-if)#exit

R3(config)#interface s1/0

R3(config-if)#ip address 172.16.13.3 255.255.255.248

R3(config-if)#no shutdown

R3(config-if)#exit

R3(config)#interface s1/1

R3(config-if)#ip address 172.16.23.3 255.255.255.248

R3(config-if)#no shutdown

R3(config-if)#exit

R3(config)#interface s1/2

R3(config-if)#ip address 172.16.34.3 255.255.255.248

R3(config-if)#no shutdown

R3(config-if)#exit

R3(config)#router eigrp 100

R3(config-router)#network 192.168.3.0

R3(config-router)#network 172.16.13.0

R3(config-router)#network 172.16.23.0

R3(config-router)#network 172.16.34.0

R3(config-router)#no auto-summary

R3#sh ip eigrp neighbors

EIGRP-IPv4 Neighbors for AS(100)

H	Address	Interface	Hold	Uptime	SRTT	RTO	Q	Seq
		(sec)	(ms)	Cnt	Num			
2	172.16.34.4	Se1/2	14	00:03:09	15	100	0	3
1	172.16.13.1	Se1/0	14	00:06:25	21	126	0	9
0	172.16.23.2	Se1/1	13	00:06:25	20	120	0	9

R3#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP

a - application route

+ - replicated route, % - next hop override

Gateway of last resort is not set

172.16.0.0/16 is variably subnetted, 7 subnets, 2 masks

D 172.16.12.0/29 [90/2681856] via 172.16.23.2, 00:16:48, Serial1/1
[90/2681856] via 172.16.13.1, 00:16:48, Serial1/0

C 172.16.13.0/29 is directly connected, Serial1/0
L 172.16.13.3/32 is directly connected, Serial1/0
C 172.16.23.0/29 is directly connected, Serial1/1
L 172.16.23.3/32 is directly connected, Serial1/1
C 172.16.34.0/29 is directly connected, Serial1/2
L 172.16.34.3/32 is directly connected, Serial1/2
D 192.168.1.0/24 [90/2297856] via 172.16.13.1, 00:16:48, Serial1/0
D 192.168.2.0/24 [90/2297856] via 172.16.23.2, 00:16:48, Serial1/1
192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.3.0/24 is directly connected, Loopback3
L 192.168.3.1/32 is directly connected, Loopback3
192.168.4.0/25 is subnetted, 2 subnets
D 192.168.4.0 [90/2297856] via 172.16.34.4, 00:13:32, Serial1/2
D 192.168.4.128 [90/2297856] via 172.16.34.4, 00:13:32, Serial1/2

R3(config)#ip access-list standard PBR-ACL

R3(config-std-nacl)#remark ACL matches R4 LAN B traffic

R3(config-std-nacl)#permit 192.168.4.128 0.0.0.127

R3(config-std-nacl)#exit

R3(config)#route-map R3-to-R1 permit

R3(config-route-map)#match ip address PBR-ACL

R3(config-route-map)#set ip next-hop 172.16.13.1

R3(config-route-map)#end

```
R3(config)#int s1/2
R3(config-if)#ip policy route-map R3-to-R1
R3(config-if)#exit
R3#sh route-map
route-map R3-to-R1, permit, sequence 10
Match clauses:
  ip address (access-lists): PBR-ACL
Set clauses:
  ip next-hop 172.16.13.1
Policy routing matches: 0 packets, 0 bytes
R3(config)#access-list 1 permit 192.168.4.0 0.0.0.255
```

R4

```
Router>enable
Router#conf t
Router(config)#hostname R4
R4(config)#interface lo4
R4(config-if)#ip address 192.168.4.1 255.255.255.128
R4(config-if)#exit
R4(config)#interface lo5
R4(config-if)#ip address 192.168.4.129 255.255.255.128
R4(config-if)#exit
R4(config)#interface s1/0
R4(config-if)#ip address 172.16.34.4 255.255.255.248
```

R4(config-if)#no shutdown

R4(config-if)#exit

R4(config)#router eigrp 100

R4(config-router)#network 192.168.4.0

R4(config-router)#network 172.16.34.0

R4(config-router)#no auto-summary

R4#sh ip eigrp neighbors

EIGRP-IPv4 Neighbors for AS(100)

H	Address	Interface (sec)	Hold Uptime (ms) Cnt	SRTT Num	RTO	Q	Seq
0	172.16.34.3	Se1/0	14 00:04:07	25	150	0	9

Before Route Maps

R4#traceroute 192.168.1.1 source 192.168.4.1

Type escape sequence to abort.

Tracing the route to 192.168.1.1

VRF info: (vrf in name/id, vrf out name/id)

1 172.16.34.3 13 msec 11 msec 10 msec

2 172.16.13.1 20 msec 17 msec *

R4#traceroute 192.168.1.1 source 192.168.4.129

Type escape sequence to abort.

Tracing the route to 192.168.1.1

VRF info: (vrf in name/id, vrf out name/id)

1 172.16.34.3 15 msec 10 msec 10 msec

2 172.16.13.1 19 msec 24 msec *

After Route Maps

R4#traceroute 192.168.1.1 source 192.168.4.1

Type escape sequence to abort.

Tracing the route to 192.168.1.1

VRF info: (vrf in name/id, vrf out name/id)

1 172.16.34.3 11 msec 10 msec 10 msec

2 172.16.13.1 21 msec 22 msec *

R4#traceroute 192.168.1.1 source 192.168.4.129

Type escape sequence to abort.

Tracing the route to 192.168.1.1

VRF info: (vrf in name/id, vrf out name/id)

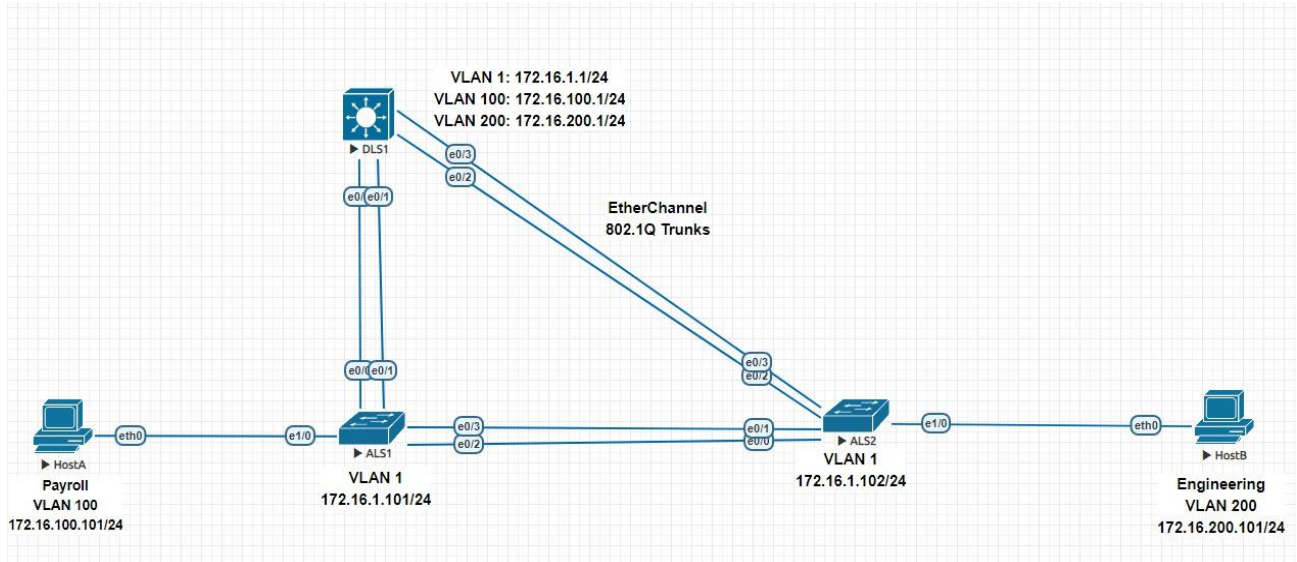
1 172.16.34.3 10 msec 10 msec 10 msec

2 172.16.13.1 18 msec 18 msec *

Practical - 6

IP Service Level Agreements and Remote SPAN in a Campus Environment

NETWORK TOPOLOGY



TASKS

- Configure trunking, VTP, and SVIs
- Implement IP SLAs to monitor various network performance characteristics
- Implement Remote Span

DLS1

Switch>en

Switch#conf t

Switch(config)#hostname DLS1

DLS1(config)#interface vlan 1

DLS1(config-if)#ip address 172.16.1.1 255.255.255.0

DLS1(config-if)#no shutdown

DLS1(config-if)#exit

Configure the trunks and EtherChannel from DLS1 to ALS1.

DLS1(config)#interface range e0/0-1

DLS1(config-if-range)#switchport trunk encapsulation dot1q

DLS1(config-if-range)#switchport mode trunk

DLS1(config-if-range)#channel-group 1 mode desirable

Creating a port-channel interface Port-channel 1

DLS1(config-if-range)#exit

Configure the trunks and EtherChannel from DLS1 to ALS2.

DLS1(config)#interface range e0/2-3

DLS1(config-if-range)#switchport trunk encapsulation dot1q

DLS1(config-if-range)#switchport mode trunk

DLS1(config-if-range)#channel-group 2 mode desirable

Creating a port-channel interface Port-channel 2

DLS1(config-if-range)#exit

Configure VTP on DLS1 and create VLANs 100 and 200 for the domain

DLS1(config)#vtp domain SWPOD

Changing VTP domain name from NULL to SWPOD

DLS1(config)#vtp version 2

DLS1(config)#vlan 100

DLS1(config-vlan)#name Payroll

DLS1(config-vlan)#exit

DLS1(config)#vlan 200

DLS1(config-vlan)#name Engineering

DLS1(config-vlan)#exit

On DLS1, create the SVIs for VLANs 100 and 200. Note that the corresponding Layer 2 VLANs must be configured for the Layer 3 SVIs to activate

DLS1(config)#interface vlan 100

DLS1(config-if)#ip address 172.16.100.1 255.255.255.0

DLS1(config-if)#no shutdown

DLS1(config-if)#exit

DLS1(config)#interface vlan 200

DLS1(config-if)#ip address 172.16.200.1 255.255.255.0

DLS1(config-if)#no shutdown

DLS1(config-if)#exit

The ip routing command is also needed to allow the DLS1 switch to act as a Layer 3 device to route between these VLANs. Because the VLANs are all considered directly connected, a routing protocol is not needed at this time. The default configuration on 3560 switches is no ip routing.

DLS1(config)#ip routing

DLS1#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP

a - application route

+ - replicated route, % - next hop override

Gateway of last resort is not set

172.16.0.0/16 is variably subnetted, 6 subnets, 2 masks

C 172.16.1.0/24 is directly connected, Vlan1

L 172.16.1.1/32 is directly connected, Vlan1

C 172.16.100.0/24 is directly connected, Vlan100

L 172.16.100.1/32 is directly connected, Vlan100

C 172.16.200.0/24 is directly connected, Vlan200

L 172.16.200.1/32 is directly connected, Vlan200

Configure the Cisco IOS IP SLA source to measure network performance

DLS1(config)#ip sla 1

DLS1(config-ip-sla)#icmp-echo 172.16.100.101

DLS1(config-ip-sla-echo)#exit

DLS1(config)#ip sla 2

DLS1(config-ip-sla)#icmp-echo 172.16.200.101

DLS1(config-ip-sla-echo)#exit

DLS1(config)#ip sla 3

DLS1(config-ip-sla)#udp-jitter 172.16.1.101 5000

DLS1(config-ip-sla-jitter)#exit

DLS1(config)#ip sla 4

DLS1(config-ip-sla)#udp-jitter 172.16.1.102 5000

DLS1(config-ip-sla-jitter)#exit

DLS1(config)#ip sla schedule 1 life forever start-time now

DLS1(config)#ip sla schedule 2 life forever start-time now

DLS1(config)#ip sla schedule 3 life forever start-time now

DLS1(config)#ip sla schedule 4 life forever start-time now

Monitor IP SLAs operations

DLS1#show ip sla configuration 1

IP SLAs Infrastructure Engine-III

Entry number: 1

Owner:

Tag:

Operation timeout (milliseconds): 5000

Type of operation to perform: icmp-echo

Target address/Source address: 172.16.100.101/0.0.0.0

Type Of Service parameter: 0x0

Request size (ARR data portion): 28

Data pattern: 0xABCDABCD

Verify data: No

Vrf Name:

Schedule:

Operation frequency (seconds): 60 (not considered if randomly scheduled)

Next Scheduled Start Time: Start Time already passed

Group Scheduled : FALSE

Randomly Scheduled : FALSE

Life (seconds): Forever

Entry Ageout (seconds): never

Recurring (Starting Everyday): FALSE

Status of entry (SNMP RowStatus): Active

Threshold (milliseconds): 5000

Distribution Statistics:

Number of statistic hours kept: 2

Number of statistic distribution buckets kept: 1

Statistic distribution interval (milliseconds): 20

Enhanced History:

History Statistics:

Number of history Lives kept: 0

Number of history Buckets kept: 15

History Filter Type: None

DLS1#show ip sla configuration 3

IP SLAs Infrastructure Engine-III

Entry number: 3

Owner:

Tag:

Operation timeout (milliseconds): 5000

Type of operation to perform: udp-jitter

Target address/Source address: 172.16.1.101/0.0.0.0

Target port/Source port: 5000/0

Type Of Service parameter: 0x0

Request size (ARR data portion): 32

Packet Interval (milliseconds)/Number of packets: 20/10

Verify data: No

Vrf Name:

Control Packets: enabled

Schedule:

Operation frequency (seconds): 60 (not considered if randomly scheduled)

Next Scheduled Start Time: Start Time already passed

Group Scheduled : FALSE

Randomly Scheduled : FALSE

Life (seconds): Forever

Entry Ageout (seconds): never

Recurring (Starting Everyday): FALSE

Status of entry (SNMP RowStatus): Active

Threshold (milliseconds): 5000

Distribution Statistics:

Number of statistic hours kept: 2

Number of statistic distribution buckets kept: 1

Statistic distribution interval (milliseconds): 20

Enhanced History:

Percentile:

DLS1#show ip sla application

IP Service Level Agreements

Version: Round Trip Time MIB 2.2.0, Infrastructure Engine-III

Supported Operation Types:

icmpEcho, path-echo, path-jitter, udpEcho, tcpConnect, http

dns, udpJitter, dhcp, ftp, lsp Group, lspPing, lspTrace

pseudowirePing, udpApp, wspApp, mcast, generic

Supported Features:

IPSLAs Event Publisher

IP SLAs low memory water mark: 225778552

Estimated system max number of entries: 165365

Estimated number of configurable operations: 165241

Number of Entries configured : 4

Number of active Entries : 4

Number of pending Entries : 0

Number of inactive Entries : 0

Time of last change in whole IP SLAs: *14:08:46.139 EET Sat Apr 11 2020

DLS1#show ip sla statistics 1

IPSLAs Latest Operation Statistics

IPSLA operation id: 1

Latest RTT: 1 milliseconds

Latest operation start time: 14:34:23 EET Sat Apr 11 2020

Latest operation return code: OK

Number of successes: 26

Number of failures: 1

Operation time to live: Forever

DLS1#show ip sla statistics 3

IPSLAs Latest Operation Statistics

IPSLA operation id: 3

Type of operation: udp-jitter

Latest RTT: 1 milliseconds

Latest operation start time: 14:34:36 EET Sat Apr 11 2020

Latest operation return code: OK

RTT Values:

Number Of RTT: 10 RTT Min/Avg/Max: 1/1/2 milliseconds

Latency one-way time:

Number of Latency one-way Samples: 6

Source to Destination Latency one way Min/Avg/Max: 0/0/1 milliseconds

Destination to Source Latency one way Min/Avg/Max: 0/0/1 milliseconds

Jitter Time:

Number of SD Jitter Samples: 9

Number of DS Jitter Samples: 9

Source to Destination Jitter Min/Avg/Max: 0/1/1 milliseconds

Destination to Source Jitter Min/Avg/Max: 0/1/1 milliseconds

Over Threshold:

Number Of RTT Over Threshold: 0 (0%)

Packet Loss Values:

Loss Source to Destination: 0

Source to Destination Loss Periods Number: 0

Source to Destination Loss Period Length Min/Max: 0/0

Source to Destination Inter Loss Period Length Min/Max: 0/0

Loss Destination to Source: 0

Destination to Source Loss Periods Number: 0

Destination to Source Loss Period Length Min/Max: 0/0

Destination to Source Inter Loss Period Length Min/Max: 0/0

Out Of Sequence: 0 Tail Drop: 0

Packet Late Arrival: 0 Packet Skipped: 0

Voice Score Values:

Calculated Planning Impairment Factor (ICPIF): 0

Mean Opinion Score (MOS): 0

Number of successes: 27

Number of failures: 0

Operation time to live: Forever

Configure Remote Span

DLS1(config)#vlan 100

DLS1(config-vlan)#remote-span

DLS1(config-vlan)#exit

DLS1(config)#monitor session 1 source interface e0/0 both

DLS1(config)# monitor session 1 destination remote vlan 100

ALS1

Switch>en

Switch#conf t

Switch(config)#hostname ALS1

ALS1(config)#interface vlan 1

ALS1(config-if)#ip address 172.16.1.101 255.255.255.0

ALS1(config-if)#no shutdown

ALS1(config-if)#exit

ALS1(config)#ip default-gateway 172.16.1.1

Configure the trunks and EtherChannel between ALS1 and DLS1

ALS1(config)#interface range e0/0-1

ALS1(config-if-range)# switchport trunk encapsulation dot1q

ALS1(config-if-range)#switchport mode trunk

ALS1(config-if-range)#channel-group 1 mode desirable

Creating a port-channel interface Port-channel 1

ALS1(config-if-range)#exit

Configure the trunks and EtherChannel between ALS1 and ALS2

ALS1(config)#interface range e0/2-3

ALS1(config-if-range)#switchport trunk encapsulation dot1q

ALS1(config-if-range)#switchport mode trunk

ALS1(config-if-range)#channel-group 2 mode desirable

Creating a port-channel interface Port-channel 2

Configure VTP on ALS1

ALS1(config)#vtp mode client

Setting device to VTP Client mode for VLANs.

ALS1(config)#int e1/0

ALS1(config-if)#switchport mode access

ALS1(config-if)#switchport access vlan 100

ALS1(config-if)#exit

Configure Cisco IOS IP SLA responders.

ALS1(config)#ip sla responder

ALS1(config)#ip sla responder udp-echo ipaddress 172.16.1.1 port 5000

ALS1#show ip sla responder

General IP SLA Responder on Control port 1967

General IP SLA Responder on Control V2 port 1167

General IP SLA Responder is: Enabled

Number of control message received: 16 Number of errors: 0

Recent sources:

172.16.1.1 [14:23:36.259 EET Sat Apr 11 2020]

172.16.1.1 [14:22:36.257 EET Sat Apr 11 2020]

172.16.1.1 [14:21:36.255 EET Sat Apr 11 2020]

172.16.1.1 [14:20:36.256 EET Sat Apr 11 2020]

172.16.1.1 [14:19:36.258 EET Sat Apr 11 2020]

Recent error sources:

Number of control v2 message received: 0 Number of errors: 0

Recent sources:

Recent error sources:

Permanent Port IP SLA Responder

Permanent Port IP SLA Responder is: Enabled

udpEcho Responder:

IP Address	Port
172.16.1.1	5000

ALS2

Switch>en

Switch#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#hostname ALS2

ALS2(config)#interface vlan 1

ALS2(config-if)#ip address 172.16.1.102 255.255.255.0

ALS2(config-if)#no shutdown

ALS2(config-if)#exit

ALS2(config)#ip default-gateway 172.16.1.1

Configure the trunks and EtherChannel between ALS2 and ALS1

ALS2(config)#interface range e0/0-1

ALS2(config-if-range)#switchport trunk encapsulation dot1q

ALS2(config-if-range)#switchport mode trunk

ALS2(config-if-range)#channel-group 2 mode desirable

Creating a port-channel interface Port-channel 2

ALS2(config-if-range)#exit

Configure the trunks and EtherChannel between ALS2 and DLS1

ALS2(config)#interface range e0/2-3

ALS2(config-if-range)#switchport trunk encapsulation dot1q

ALS2(config-if-range)#switchport mode trunk

ALS2(config-if-range)#channel-group 1 mode desirable

Creating a port-channel interface Port-channel 1

ALS2(config-if-range)#exit

Configure VTP on ALS2

ALS2(config)#vtp mode client

Setting device to VTP Client mode for VLANs

ALS2(config)#int e1/0

ALS2(config-if)#switchport mode access

ALS2(config-if)#switchport access vlan 200

ALS2(config-if)#exit

Configure Cisco IOS IP SLA responders.

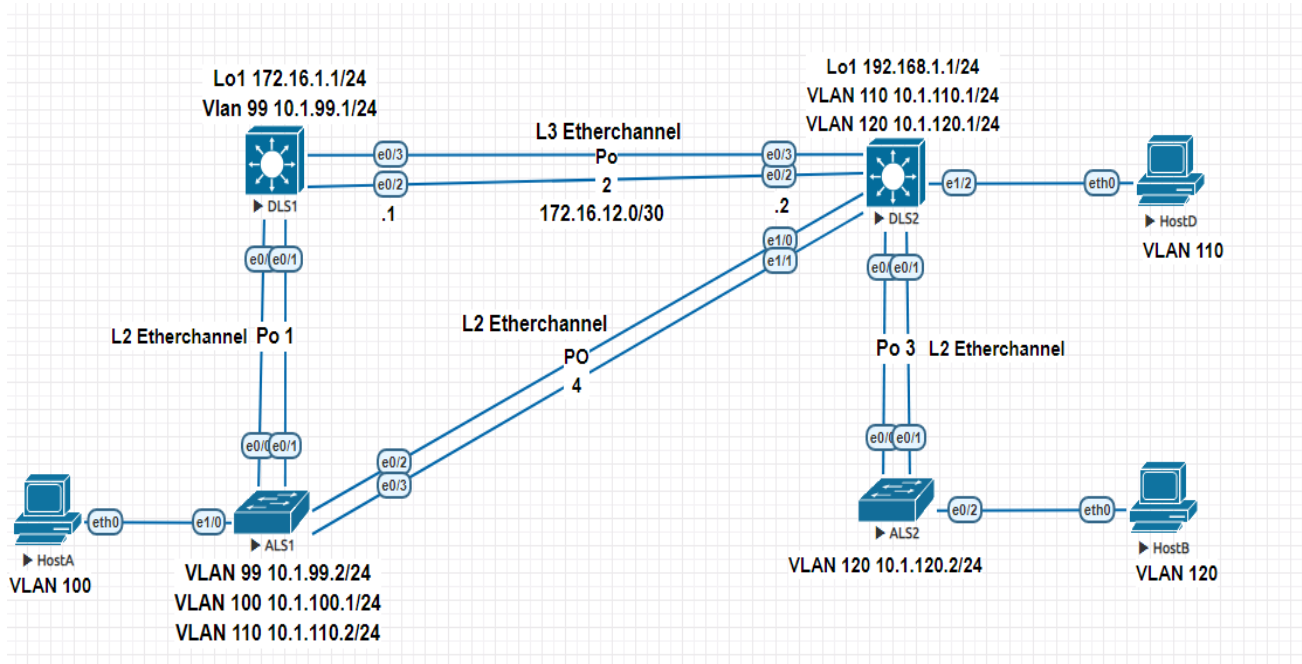
ALS2(config)#ip sla responder

ALS2(config)#ip sla responder udp-echo ipaddress 172.16.1.1 port 5000

Practical - 7

Inter-VLAN Routing

NETWORK TOPOLOGY



TASKS

- Implement a Layer 3 EtherChannel
- Implement Static Routing
- Implement Inter-Vlan Routing

DLS1

Switch>enable

Switch#conf t

Switch(config)#hostname DLS1

DLS1(config)#interface loopback 1

DLS1(config-if)#ip address 172.16.1.1 255.255.255.0

DLS1(config-if)#exit

DLS1(config)#interface vlan 99

DLS1(config-if)#ip address 10.1.99.1 255.255.255.0

DLS1(config-if)#no shutdown

Implement a Layer 3 EtherChannel

DLS1(config)#int range e0/2-3

DLS1(config-if-range)#no switchport

DLS1(config-if-range)#no ip address

DLS1(config-if-range)#channel-group 2 mode on

Creating a port-channel interface Port-channel 2

DLS1(config-if-range)#exit

DLS1(config)#interface port-channel 2

DLS1(config-if)#ip address 172.16.12.1 255.255.255.252

DLS1(config-if)#end

DLS1(config)#int range e0/0-1

DLS1(config-if-range)#switchport trunk encapsulation dot1q

DLS1(config-if-range)#switchport mode trunk

DLS1(config-if-range)#channel-group 1 mode desirable

Creating a port-channel interface Port-channel 1

DLS1(config-if-range)#end

DLS1#sh interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Po1	on	802.1q	trunking	1

Port Vlan allowed on trunk

Po1 1-4094

Port Vlan allowed and active in management domain

Po1 1,99

Port Vlan in spanning tree forwarding state and not pruned

Po1 1,99

Implement Static Routing

DLS1(config)#ip routing

DLS1(config)#ip route 192.168.1.0 255.255.255.252 172.16.12.2

DLS1(config)# ip route 192.168.1.0 255.255.255.0 10.1.120.1

DLS1(config)# ip route 192.168.1.0 255.255.255.0 10.1.110.1

DLS1#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP

a - application route

+ - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

C 10.1.99.0/24 is directly connected, Vlan99

L 10.1.99.1/32 is directly connected, Vlan99

172.16.0.0/16 is variably subnetted, 4 subnets, 3 masks

C 172.16.1.0/24 is directly connected, Loopback1

L 172.16.1.1/32 is directly connected, Loopback1
C 172.16.12.0/30 is directly connected, Port-channel2
L 172.16.12.1/32 is directly connected, Port-channel2
192.168.1.0/30 is subnetted, 1 subnets
S 192.168.1.0 [1/0] via 172.16.12.2

DLS2

Switch>en

Switch#conf t

Switch(config)#hostname DLS2

DLS2(config)#interface loopback 1

DLS2(config-if)#ip address 192.168.1.1 255.255.255.0

DLS2(config-if)#exit

DLS2(config)#interface vlan 110

DLS2(config-if)#ip address 10.1.110.1 255.255.255.0

DLS2(config-if)#no shutdown

DLS2(config-if)#exit

DLS2(config)#interface vlan 120

DLS2(config-if)#ip address 10.1.120.1 255.255.255.0

DLS2(config-if)#no shutdown

DLS2(config-if)#exit

Implement a Layer 3 EtherChannel

DLS2(config)#interface range e0/2-3

DLS2(config-if-range)#no switchport

DLS2(config-if-range)#no ip

DLS2(config-if-range)#no ip address

DLS2(config-if-range)#channel-group 2 mode on

Creating a port-channel interface Port-channel 2

DLS2(config-if-range)#exit

DLS2(config)#interface port-channel 2

DLS2(config-if)#ip address 172.16.12.2 255.255.255.252

DLS2(config-if)#end

DLS2(config)#interface range e0/0-1

DLS2(config-if-range)#switchport trunk encapsulation dot1q

DLS2(config-if-range)#switchport mode trunk

DLS2(config-if-range)#channel-group 3 mode desirable

Creating a port-channel interface Port-channel 3

DLS2(config-if-range)#exit

DLS2(config)#interface range e1/0-1

DLS2(config-if-range)#switchport trunk encapsulation dot1q

DLS2(config-if-range)#switchport mode trunk

DLS2(config-if-range)#channel-group 4 mode desirable

Creating a port-channel interface Port-channel 4

DLS2(config-if-range)#end

DLS2#sh interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Po3	on	802.1q	trunking	1
Po4	on	802.1q	trunking	1

Port Vlan allowed on trunk

Po3 1-4094

Po4 1-4094

Port Vlan allowed and active in management domain

Po3 1,110,120

Po4 1,110,120

Port Vlan in spanning tree forwarding state and not pruned

Po3 1,110,120

Po4 1,110,120

Implement Static Routing

DLS2(config)#ip routing

DLS2(config)#ip route 172.16.1.0 255.255.255.252 172.16.12.1

DLS2(config)# ip route 172.16.1.0 255.255.255.0 10.1.99.1

Configure the host ports for the appropriate VLANs according to the diagram

DLS2(config)#interface e1/2

DLS2(config-if)#switchport mode access

DLS2(config-if)#switchport access vlan 110

DLS2#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP

a - application route

+ - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks

C 10.1.110.0/24 is directly connected, Vlan110

L 10.1.110.1/32 is directly connected, Vlan110

C 10.1.120.0/24 is directly connected, Vlan120

L 10.1.120.1/32 is directly connected, Vlan120

172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks

S 172.16.1.0/30 [1/0] via 172.16.12.1
C 172.16.12.0/30 is directly connected, Port-channel2
L 172.16.12.2/32 is directly connected, Port-channel2
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.1.0/24 is directly connected, Loopback1
L 192.168.1.1/32 is directly connected, Loopback1

ALS1

Switch>en

Switch#conf t

Switch(config)#hostname ALS1

ALS1(config)#ip default-gateway 10.1.99.1

ALS1(config)#ip default-gateway 10.1.110.1

ALS1(config)#ip default-gateway 10.1.100.2

Implement a Layer 3 EtherChannel

ALS1(config)#int range e0/0-1

ALS1(config-if-range)#switchport trunk encapsulation dot1q

ALS1(config-if-range)#switchport mode trunk

ALS1(config-if-range)#channel-group 1 mode desirable

Creating a port-channel interface Port-channel 1

ALS1(config-if-range)#exit

ALS1(config)#int range e0/2-3

ALS1(config-if-range)#switchport trunk encapsulation dot1q

ALS1(config-if-range)#switchport mode trunk

ALS1(config-if-range)#channel-group 4 mode desirable

Creating a port-channel interface Port-channel 4

ALS1(config-if-range)#end

ALS1#sh etherchannel summary

Flags: D - down P - bundled in port-channel

I - stand-alone s - suspended

H - Hot-standby (LACP only)

R - Layer3 S - Layer2

U - in use N - not in use, no aggregation

f - failed to allocate aggregator

M - not in use, minimum links not met

m - not in use, port not aggregated due to minimum links not met

u - unsuitable for bundling

w - waiting to be aggregated

d - default port

A - formed by Auto LAG

Number of channel-groups in use: 2

Number of aggregators: 2

Group Port-channel Protocol Ports

-----+-----+-----+-----

1 Po1(SU) PAgP Et0/0(P) Et0/1(P)

4 Po4(SU) PAgP Et0/2(P) Et0/3(P)

Configure the host ports for the appropriate VLANs according to the diagram

ALS1(config)#interface e1/0

ALS1(config-if)#switchport mode access

ALS1(config-if)#switchport access vlan 100

ALS2

Switch>en

Switch#conf t

Switch(config)#hostname ALS2

ALS2(config)#ip default-gateway 10.1.120.1

Implement a Layer 3 EtherChannel

ALS2(config)#int range e0/0-1

ALS2(config-if-range)#switchport trunk encapsulation dot1q

ALS2(config-if-range)#switchport mode trunk

ALS2(config-if-range)#channel-group 3 mode desirable

Creating a port-channel interface Port-channel 3

ALS2(config-if-range)#end

ALS2#sh etherchannel summary

Flags: D - down P - bundled in port-channel

I - stand-alone s - suspended

H - Hot-standby (LACP only)

R - Layer3 S - Layer2

U - in use N - not in use, no aggregation

f - failed to allocate aggregator

M - not in use, minimum links not met

m - not in use, port not aggregated due to minimum links not met

u - unsuitable for bundling

w - waiting to be aggregated

d - default port

A - formed by Auto LAG

Number of channel-groups in use: 1

Number of aggregators: 1

Group Port-channel Protocol Ports

```
-----+-----+-----+-----+-----+
3   Po3(SU)   PAgP   Et0/0(P) Et0/1(P)
```

Configure the host ports for the appropriate VLANs according to the diagram

ALS2(config)#interface e0/2

ALS2(config-if)#switchport mode access

ALS2(config-if)#switchport access vlan 120

HOST A

VPCS> ip 10.1.100.1 255.255.255.0 10.1.100.2

HOST B

VPCS> ip 10.1.120.2 255.255.255.0 10.1.120.1

HOST D

VPCS> ip 10.1.110.2 255.255.255.0 10.1.110.1

